CSEC Mathematics

January 2010 – Paper 2

Solutions

Kerwin Springer

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SECTION I

Answer ALL questions in this section.

All working must be clearly shown.

1. (a) Using a calculator, or otherwise, calculate the exact value of

$$\frac{2.76}{0.8} + 8.7^2$$ [3]

Using a calculator,

$$\frac{2.76}{0.8} + 8.7^2 = 3.45 + 75.69$$

$$= 79.14$$

(b) In a certain company, a salesman is paid a fixed salary of \$3 140 per month plus an annual commission of 2% on the TOTAL value of cars sold for the year. If the salesman sold cars valued at \$720 000 in 2009, calculate

(i) his fixed salary for the year [1]

Fixed salary for the year = Monthly salary × 12

$= \$3140 \times 12$

$= \$37\ 680$

(ii) the amount he received in commission for the year [1]

Amount in commission = 2% of \$720 000

$= \frac{2}{100} \times \$720\ 000$

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$= \$14\ 400$

(iii) his TOTAL income for the year [1]

Total income = Fixed salary + Amount earned in commission

$= \$37\ 680 + \$14\ 400$

$= \$52\ 080$

∴ His total income for the year is \$52 080.

(c) The ingredients for making pancakes are shown in the diagram below.

Ingredients for making 8 pancakes
2 cups pancake mix $1\frac{1}{3}$ cups milk

(i) Ryan wishes to make 12 pancakes using the instructions given above.

Calculate the number of cups of pancake mix he must use. [2]

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We are given that 8 pancakes are produced from using 2 cups of mix.

So, we have,

8 pancakes = 2 cups of mix

1 pancake = $\frac{2}{8}$ cups of mix

12 pancakes = $12 \times \frac{2}{8}$

= 3 cups of mix

∴ He must use 3 cups of pancake mix.

(ii) Neisha used 5 cups of milk to make pancakes using the same instructions. How many pancakes did she make? [3]

We are given that $1\frac{1}{3}$ cups of milk produce 8 pancakes.

So, we have,

$1\frac{1}{3}$ cups of milk = 8 pancakes

1 cup of milk = $\frac{8}{1\frac{1}{3}}$

= 6 pancakes

5 cups of milk = 5×6

= 30 pancakes

∴ Neisha made 30 pancakes.

Total: 11 marks

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2. (a) Given that $a = 6, b = -4$ and $c = 8$, calculate the value of [3]

$$\frac{a^2+b}{c-b}$$

We are given that $a = 6, b = -4$ and $c = 8$.

$$\frac{a^2+b}{c-b} = \frac{(6)^2+(-4)}{(8)-(-4)}$$

$$= \frac{36-4}{8+4}$$

$$= \frac{32}{12}$$

$$= \frac{8}{3}$$

(b) Simplify the expression:

(i) $3(x-y)+4(x+2y)$ [2]

$$3(x-y)+4(x+2y)$$

$$= 3x-3y+4x+8y$$

$$= 7x+5y$$

(ii) $\frac{4x^2 \times 3x^4}{6x^3}$ [3]

$$\frac{4x^2 \times 3x^4}{6x^3}$$

$$= \frac{12x^6}{6x^3}$$

$$= 2x^3$$

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(c) (i) Solve the inequality

$$x - 3 < 3x - 7$$ [3]

$x - 3 < 3x - 7$

$7 - 3 < 3x - x$

$4 < 2x$

$2x > 4$

$x > \frac{4}{2}$

$x > 2$

(ii) If x is an integer, determine the SMALLEST value of x that satisfies the inequality in (c)(i) above. [1]

$x > 2$ and $x \in \mathbb{Z}$.

$\mathbb{Z} = \{\ldots - 3, -2, -2, 0, 1, 2, 3, \ldots\}$

$\therefore$ The smallest integer that satisfies the inequality is $x = 3$.

Total: 12 marks

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3. (a) T and E are subsets of a universal set, U, such that:

$U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$

$T = \{\text{multiples of } 3\}$

$E = \{\text{even numbers}\}$

(i) Draw a Venn diagram to represent this information. [4]

$U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$

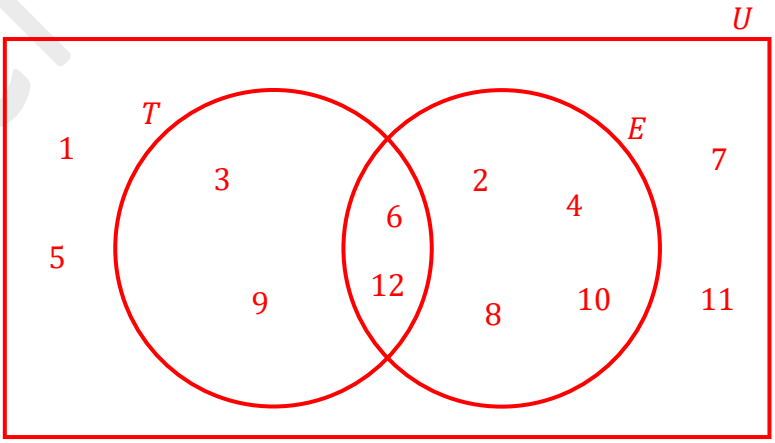
$T = \{multiples\ of\ 3\}$

$= \{3, 6, 9, 12\}$

$E = \{even\ numbers\}$

$= \{2, 4, 6, 8, 10, 12\}$

The Venn diagram is as follows:



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(ii) List the members of the set

(a) $T \cap E$ [1]

$T \cap E = \{6, 12\}$

(b) $(T \cup E)'$ [1]

$(T \cup E)' = \{1, 5, 7, 11\}$

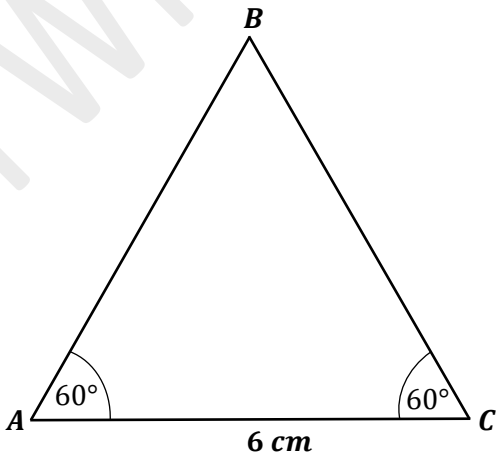
(b) **Using a pencil, a ruler, and a pair of compasses only:**

(i) Construct, accurately, the triangle ABC shown below, where,

$AC = 6\ cm$

$\angle ACB = 60°$

$\angle CAB = 60°$ [3]



Required to construct accurately, the triangle ABC, where

$AC = 6\ cm, \angle ACB = 60°$ and $\angle CAB = 60°$.

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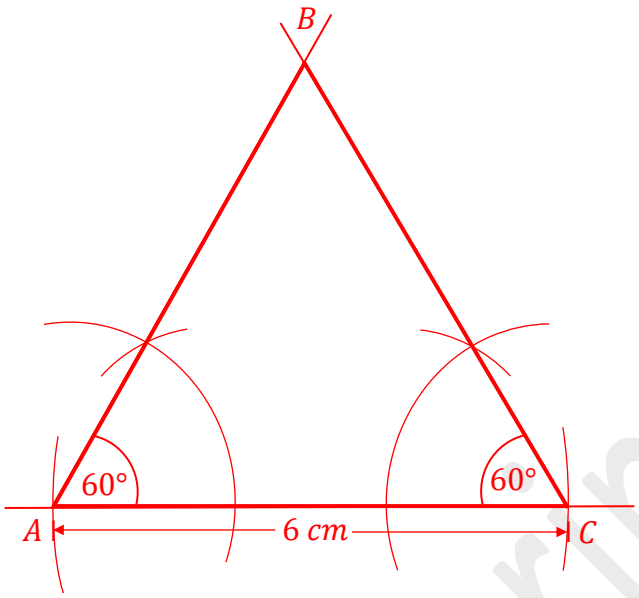
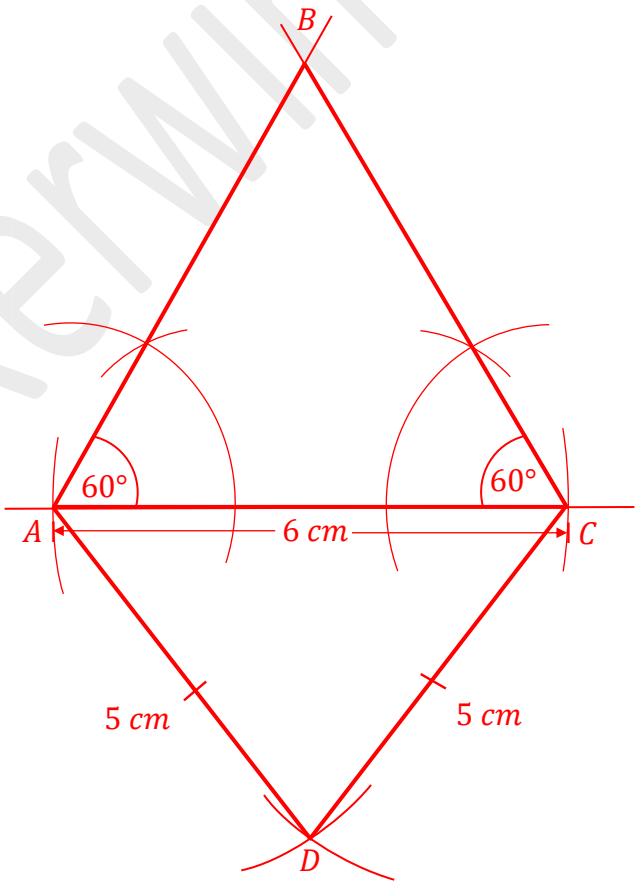
The diagram is shown as follows:



(ii) Complete the diagram to show the kite, $ABCD$, in which

$AD = 5\ cm$. [2]



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(iii) Measure and state the size of $\angle DAC$. [1]

By measurement, $D\hat{A}C = 53°$.

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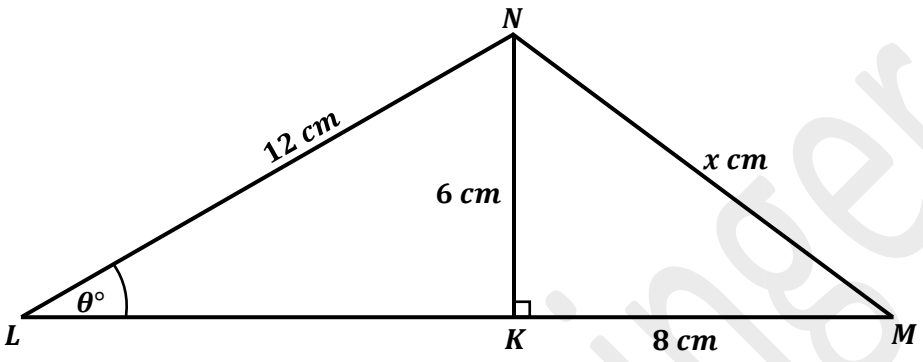
4. (a) The diagram below, **not drawn to scale**, shows a triangle LMN with $LN = 12$ cm, $NM = x\ cm$, and $\angle NLM = \theta°$. The point K on LM is such that NK is perpendicular to LM, $NK = 6\ cm$, and $KM = 8\ cm$.



Calculate the value of

(i) x [2]

Consider triangle NKM.

By Pythagoras' Theorem,

$c^2 = a^2 + b^2$

$x^2 = (6)^2 + (8)^2$

$x^2 = 36 + 64$

$x^2 = 100$

$x = \sqrt{100}$

$x = 10\ cm$

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(ii) θ [3]

Consider triangle NLK.

$\sin\theta = \frac{opp}{hyp}$

$\sin\theta = \frac{6}{12}$

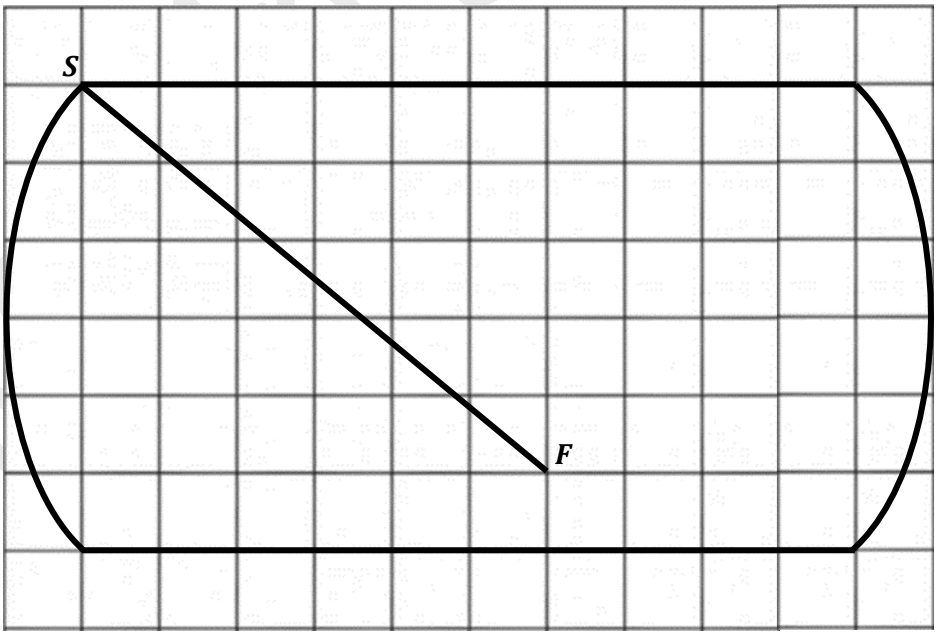
$\sin\theta = \frac{1}{2}$

$\theta = \sin^{-1}\left(\frac{1}{2}\right)$

$\theta = 30°$

(b) The diagram below shows a map of a playing field drawn on a grid of 1 cm squares.

The scale of the map is 1 : 1 250.



(i) Measure and state, in centimetres, the distance from S to F on the map. [1]

By measurement, $SF = 7.8\ cm$.

(ii) Calculate the distance, in metres, from S to F on the ACTUAL playing field. [2]

The scale is 1 : 1 250.

$1\ cm$ on the map represents $1250\ cm$ on the field.

$7.8\ cm$ on the map $= 7.8 \times 1250$

$= 9750\ cm$ on the field.

Now,

$100\ cm = 1\ m$

$1\ cm = \frac{1}{100}$

$9750\ cm = \frac{1}{100} \times 9750$

$= 97.5\ m$

$\therefore$ The distance from S to F on the actual playing field is $97.5\ m$.

(iii) Daniel ran the distance from S to F in 9.72 seconds. Calculate his average speed in

(a) m/s

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$$\text{Average speed} = \frac{Distance\ covered}{Time\ taken}$$

$$= \frac{97.5}{9.72}$$

$$= 10.0\ m/s \qquad \text{(to 3 significant figures)}$$

(b) km/h

$$\text{Distance} = 97.5\ m$$

$$= 97.5 \div 1000\ km$$

$$= 0.0975\ km$$

Now,

$$1 \text{ hour} = 60 \times 60$$

$$1 \text{ hour} = 3600 \text{ seconds}$$

$$3600 \text{ seconds} = 1 \text{ hour}$$

$$1 \text{ second} = \frac{1}{3600}$$

$$9.72 \text{ seconds} = 0.0027 \text{ hours}$$

$$\text{Time} = 9.72 \text{ seconds}$$

$$= 0.0027 \text{ hours}$$

$$\text{Average speed} = \frac{Distance\ covered}{Time\ taken}$$

$$= \frac{0.0975}{0.0027}$$

$$= 36.1\ km/h \qquad \text{(to 3 significant figures)}$$

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giving your answer correct to 3 significant figures. [3]

Total: 11 marks

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5. (a) A straight line passes through the point $T(4, 1)$ and has a gradient of $\frac{3}{5}$.

Determine the equation of this line. [3]

Substituting $m = \frac{3}{5}$ and point $T(4, 1)$ into $y - y_1 = m(x - x_1)$ gives:

$$y - 1 = \frac{3}{5}(x - 4)$$

$$y - 1 = \frac{3}{5}x - \frac{12}{5}$$

$$y = \frac{3}{5}x - \frac{12}{5} + 1$$

$$y = \frac{3}{5}x - \frac{7}{5}$$

$$5y = 3x - 7$$

$\therefore$ The equation of the line is: $5y = 3x - 7$

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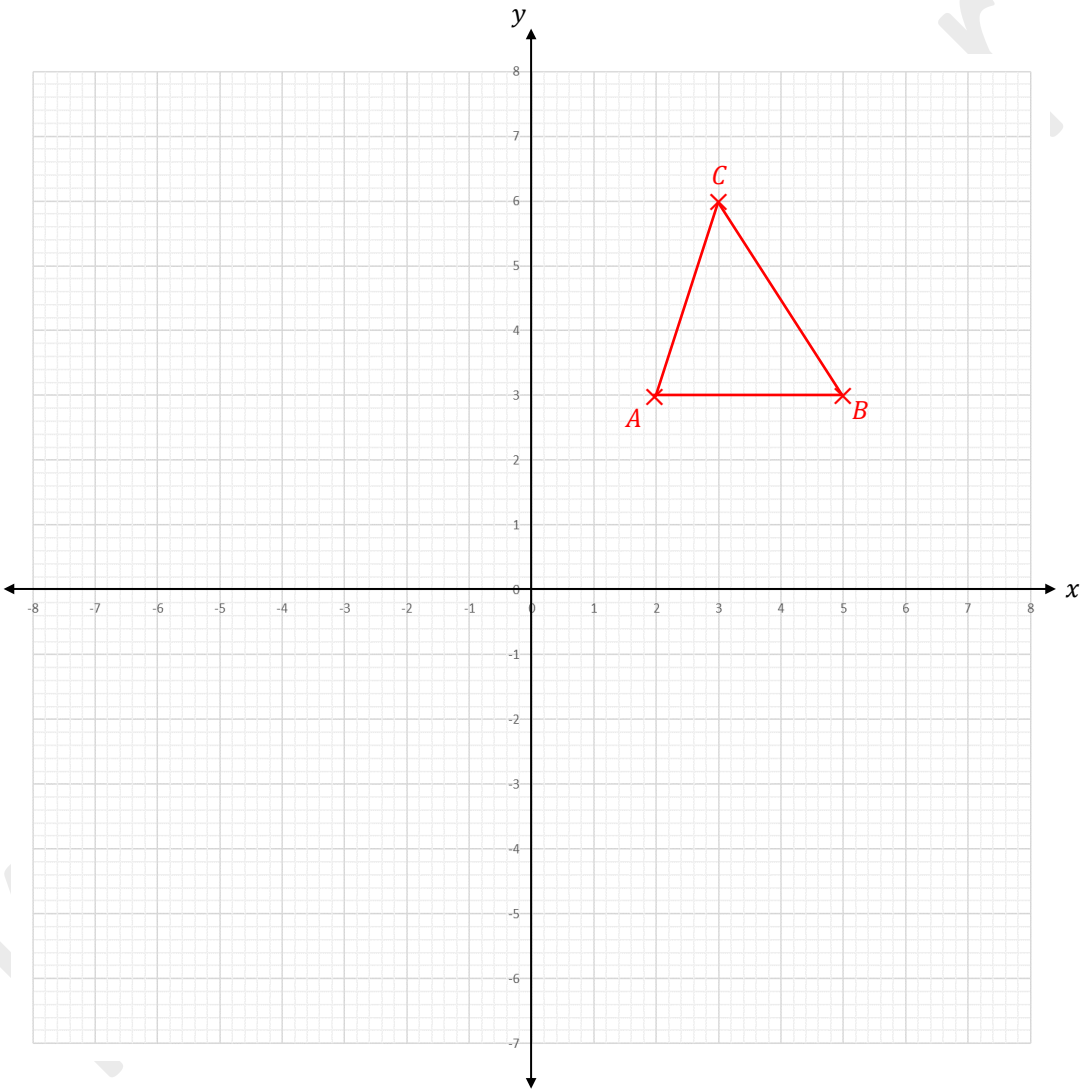
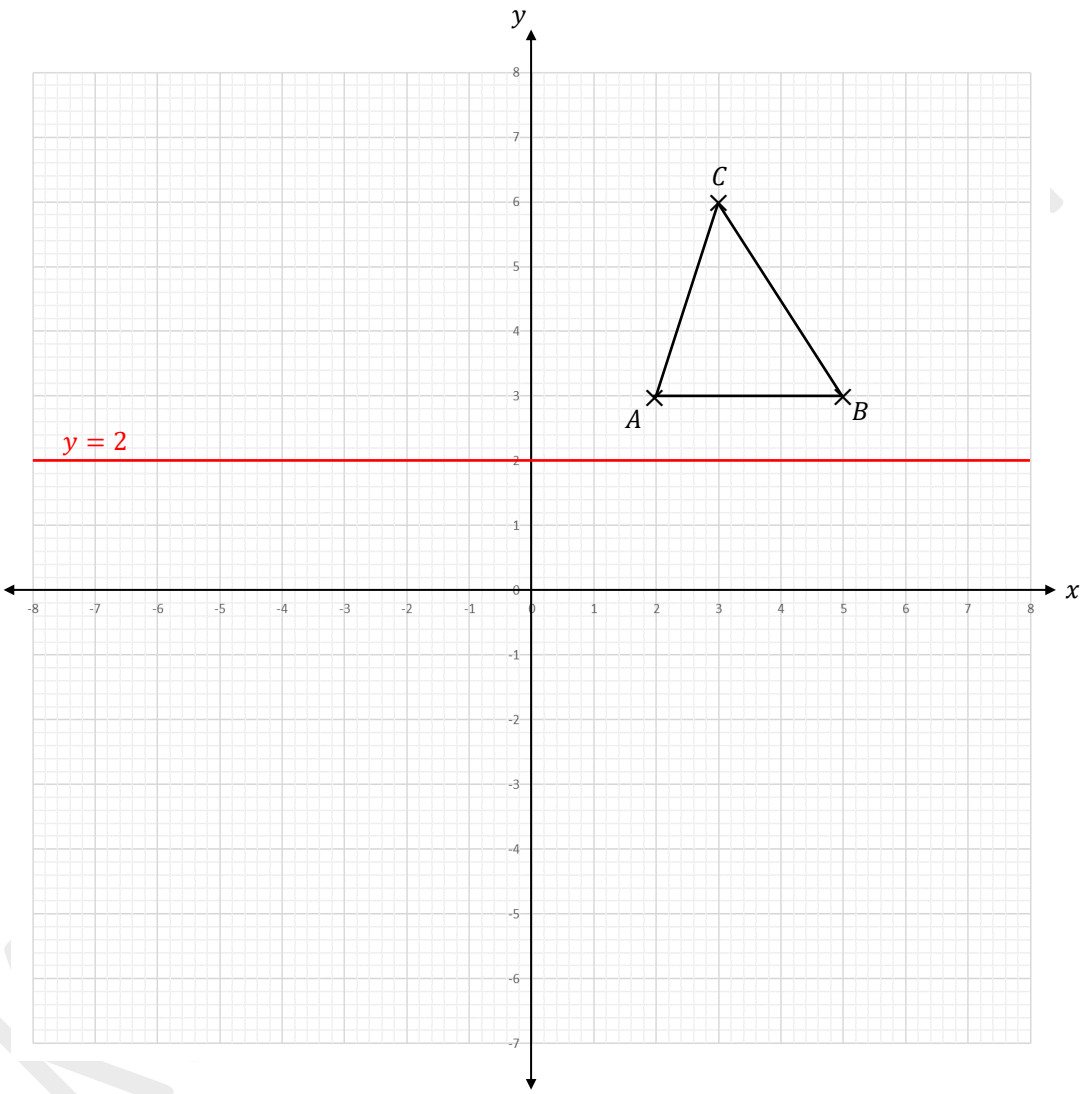
(b) (i) **Using a scale of 1 cm to represent 1 unit on both axes**, draw the triangle ABC with vertices $A(2,3)$, $B(5,3)$ and $C(3,6)$. [3]

The graph is shown below.



(ii) On the same axes used in (b)(i), draw and label the line $y = 2$. [1]

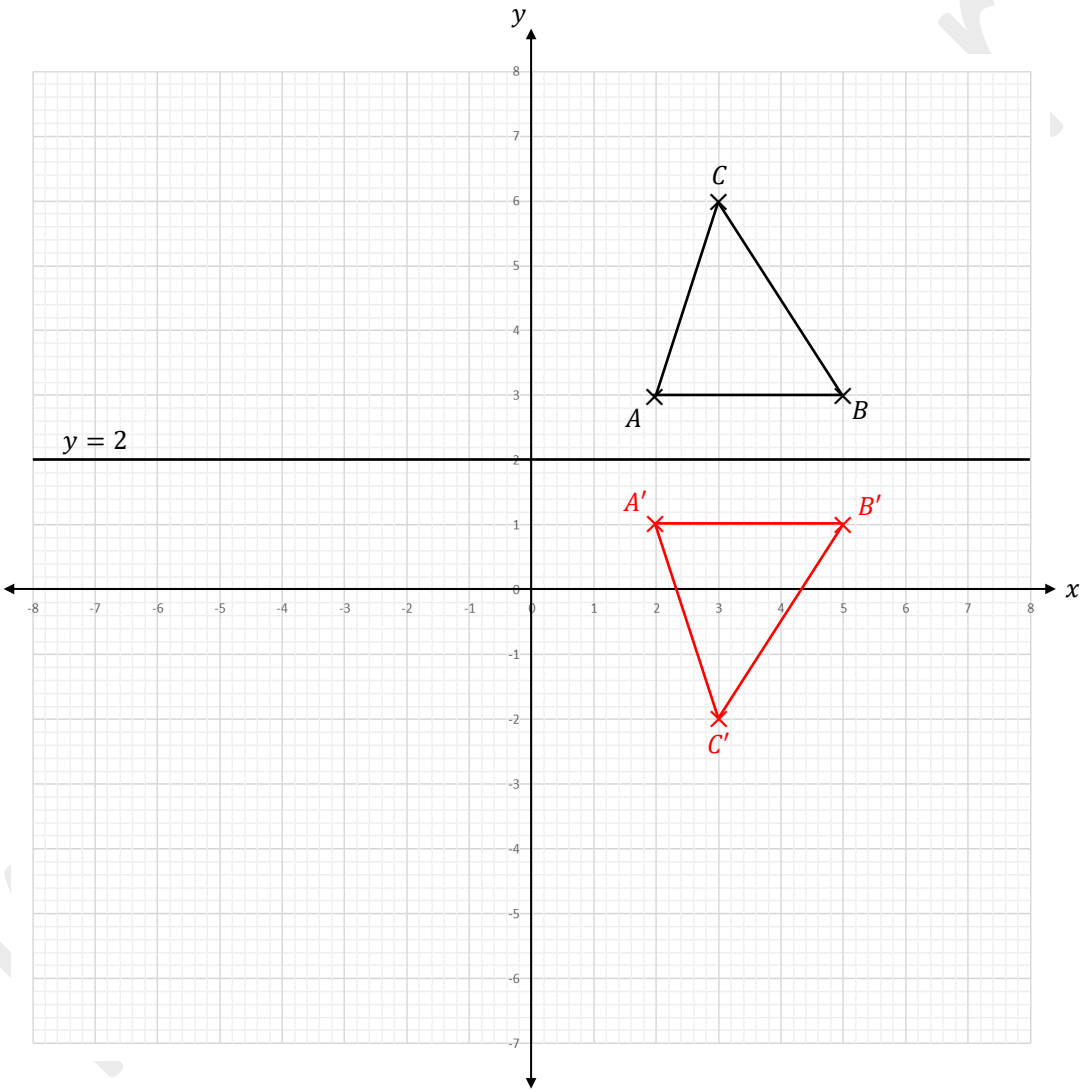
The graph is shown below.



(iii) Draw the image of triangle ABC under a reflection in the line $y = 2$.

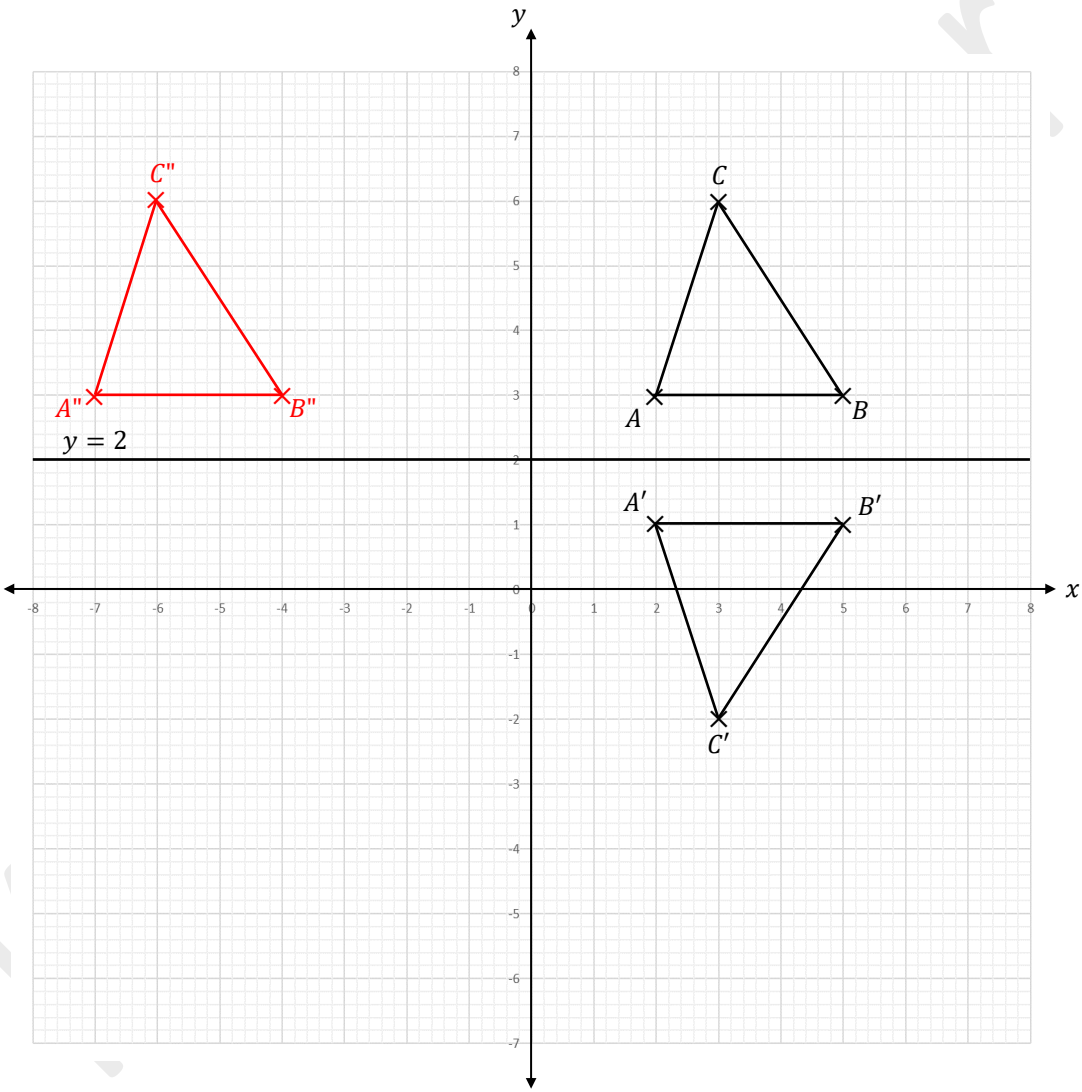
Label the image $A'B'C'$. [2]

The graph is shown below.



(iv) Draw a new triangle $A"B"C"$ with vertices $A"(-7, 4)$, $B"(-4, 4)$ and $C"(-6, 7)$. [1]

The graph is shown below.



(v) Name and describe the single transformation that maps triangle ABC onto triangle $A"B"C"$. [2]

The transformation is a translation.

Consider points $A(2, 3)$ and $A"(-7, 4)$.

The point A is mapped onto $A"$ by a horizontal shift of 9 units to the left and a vertical shift of 1 unit upwards.

The translation is represented by $T = \begin{pmatrix} -9 \\ 1 \end{pmatrix}$.

$\therefore \Delta ABC$ is mapped into $\Delta A"B"C"$ by a translation, $T = \begin{pmatrix} -9 \\ 1 \end{pmatrix}$.

Total: 12 marks

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6. A class of 26 students each recorded the distance travelled to school. The distance, to the nearest km, is recorded below:

21	11	3	22	6	32	22	18	28
26	16	17	34	12	25	8	19	14
39	17	22	24	30	18	13	23	

(a) Copy and complete the frequency table to represent this data. [2]

Distance in kilometres	Frequency
1 – 5	1
6 – 10	2
11 – 15	4
16 – 20	6
21 – 25	7
26 – 30	3
31 – 35	2
36 – 40	1

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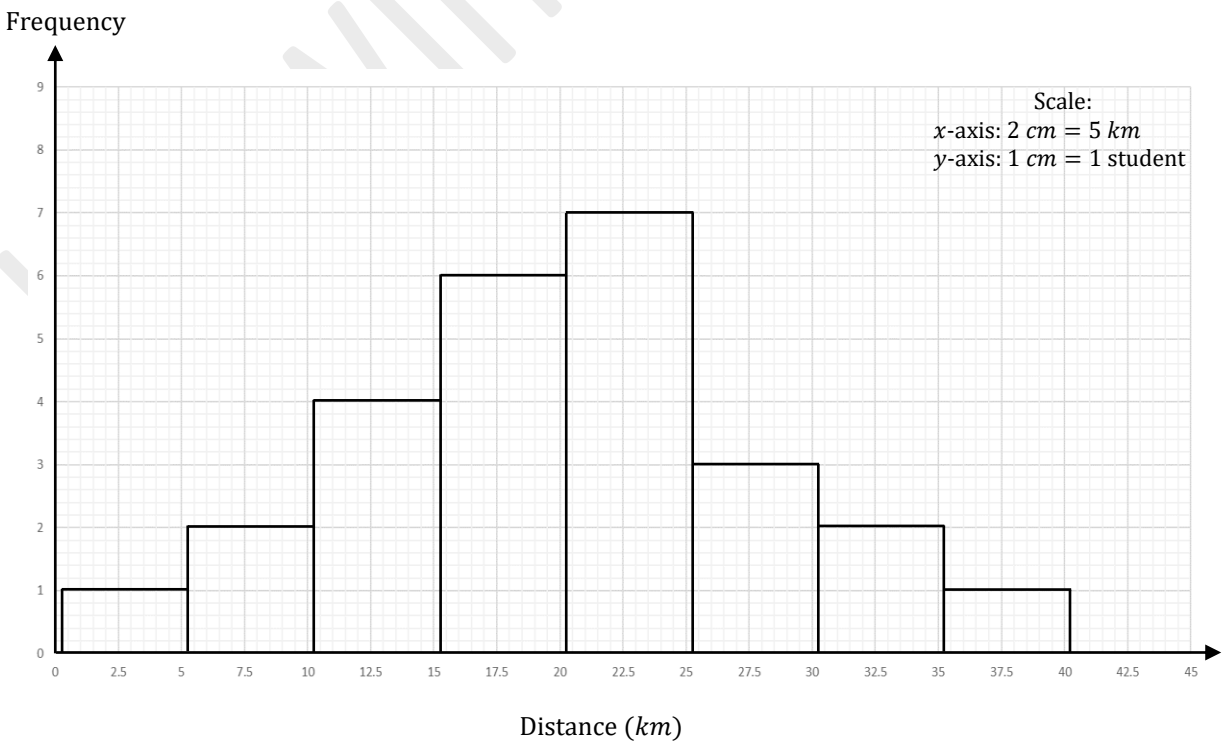
(b) **Using a scale of 2 *cm* to represent 5 *km*** on the horizontal axis and **a scale of 1 *cm* to represent 1 student** on the vertical axis, draw a histogram to represent the data. [5]

Distance in kilometres	LCB	UCB	Frequency
1 – 5	0.5	5.5	1
6 – 10	5.5	10.5	2
11 – 15	10.5	15.5	4
16 – 20	15.5	20.5	6
21 – 25	20.5	25.5	7
26 – 30	25.5	30.5	3
31 – 35	30.5	35.5	2
36 – 40	35.5	40.5	1

Title: Graph showing a histogram representing the data given.



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(c) Calculate the probability that a student chosen at random from this class recorded the distance travelled to school as 26 km or **more**. [2]

$$P(student\ travelled \geq 26\ km) = \frac{\text{Number of students travelling} \geq 26\text{ km}}{\text{Total number of students}}$$

$$P(student\ travelled \geq 26\ km) = \frac{3+2+1}{26}$$

$$P(student\ travelled \geq 26\ km) = \frac{6}{26}$$

$$P(student\ travelled \geq 26\ km) = \frac{3}{13}$$

(d) The P.T.A. plans to set up a transportation service for the school. Which average, mean, more or median, is MOST appropriate for estimating the cost of the service? Give a reason for your answer. [2]

The mean is most appropriate for estimating the cost of transportation because it takes into account each student and the actual distance that is covered.

Total: 11 marks

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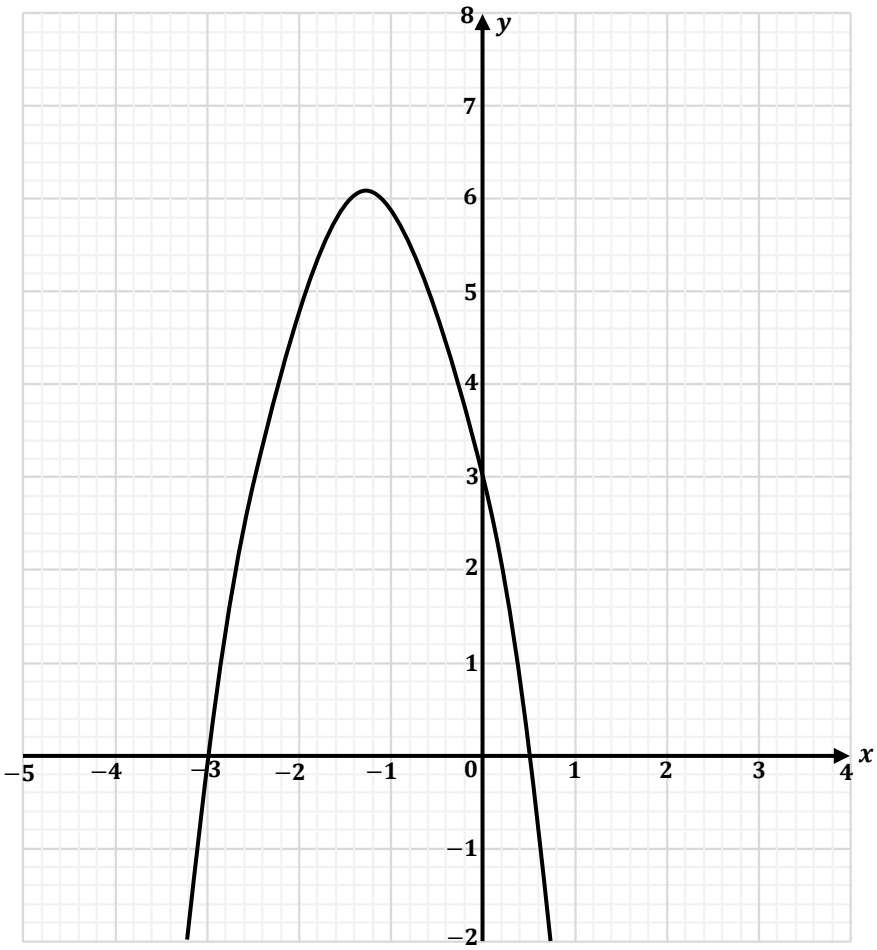
7. The graph shown below represents a function of the form: $f(x) = ax^2 + bx + c$



Using the graph above, determine

(i) the value of $f(x)$ when $x = 0$ [1]

From graph, when $x = 0, f(x) = 3$.

(ii) the values of x when $f(x) = 0$ [2]

From graph, when $f(x) = 0, x = -3$ and $x = \frac{1}{2}$.

These are the points where the curve cuts the x-axis.

(iii) the coordinates of the maximum point [2]

The coordinates of the maximum point are $(-1.3, 6.1)$.

(iv) the equation of the axis of symmetry [2]

The equation of the axis of symmetry is: $x = -1.3$.

(v) the values of x when $f(x) = 5$ [2]

From graph, when $f(x) = 5, x = -2$ and $x = -\frac{1}{2}$.

(vi) the interval within which x lies when $f(x) > 5$. Write your answer in the form $a < x < b$. [2]

The range of values of x for which $f(x)$ lies above the line $y = 5$ is $\{x: -2 < x < -0.5\}$.

Total: 11 marks

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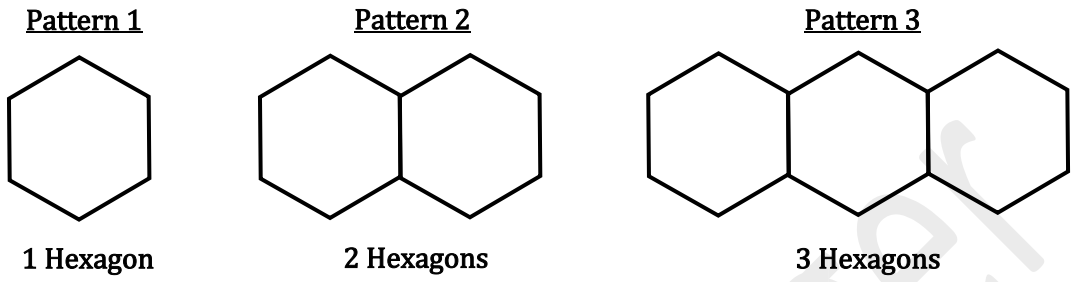
8. Bianca makes hexagons using sticks of equal length. She then creates patterns by joining the hexagons togather. Patterns 1, 2 and 3 are shown below.



The table below shows the number of hexagons in EACH pattern created and the number of sticks used to make EACH pattern.

Number of hexagons in the pattern	1	2	3	4	5	20	n
Number of sticks used for the pattern	6	11	16	x	y	z	S

(a) Determine the values of

(i) x [2]

For n hexagons in the pattern, the number of sticks used is $S = 5n + 1$.

When $n = 4$ and $S = x$,

$x = 5(4) + 1$

$= 20 + 1$

$= 21$

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(ii) y [2]

When $n = 5$ and $S = y$,

$y = 5(5) + 1$

$= 25 + 1$

$= 26$

(iii) z [2]

When $n = 20$ and $S = z$,

$z = 5(20) + 1$

$= 100 + 1$

$= 101$

(b) Write down an expression for $\boldsymbol{S}$ in terms of $\boldsymbol{n}$, where $\boldsymbol{S}$ represents the number of sticks used to make a pattern of $\boldsymbol{n}$ hexagons. [2]

S represents the number of sticks used to make a pattern of n hexagons.

$\therefore S = 5n + 1$

(c) Bianca used a total of 76 sticks to make a pattern of $\boldsymbol{h}$ hexagons. Determine the value of $\boldsymbol{h}$. [2]

Bianca used a total of 76 sticks to make a pattern of h hexagons.

When $n = h$ and $S = 76$,

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$$76 = 5(h) + 1$$

$$76 - 1 = 5h$$

$$75 = 5h$$

$$\frac{75}{5} = h$$

$$15 = h$$

Total: 10 marks

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STUDENT HUB

SECTION II

Answer TWO questions in this section.

ALGEBRA AND RELATIONS, FUNCTIONS AND GRAPHS

9. (a) The relationship between kinetic energy, E, mass, m, and velocity, v, for a moving particle is

$$E = \frac{1}{2}mv^2$$

(i) Express v in terms of E and m. [3]

$$E = \frac{1}{2}mv^2$$

$$2E = mv^2$$

$$\frac{2E}{m} = v^2$$

$$\sqrt{\frac{2E}{m}} = v$$

$$\therefore v = \sqrt{\frac{2E}{m}}$$

(ii) Determine the value of v when $E = 45$ and $m = 13$. [2]

Substituting $E = 45$ and $m = 13$ into $v = \sqrt{\frac{2E}{m}}$ gives:

$$v = \sqrt{\frac{2(45)}{13}}$$

$$= \sqrt{\frac{90}{13}}$$

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$= 2.63$ (to 2 decimal places)

(b) Given $g(x) = 3x^2 - 8x + 2$,

(i) write $g(x)$ in the form $a(x+b)^2 + c$, where a, b and $c \in \mathbb{R}$ [3]

$g(x) = 3x^2 - 8x + 2$ is in the form $ax^2 + bx + c$,

where $a = 3, b = -8$ and $c = 2$.

$$h = \frac{b}{2a}$$
$$= \frac{-8}{2(3)}$$
$$= \frac{-8}{6}$$
$$= -\frac{4}{3}$$

$$k = \frac{4ac - b^2}{4a}$$
$$= \frac{4(3)(2) - (-8)^2}{4(3)}$$
$$= \frac{24 - 64}{12}$$
$$= \frac{-40}{12}$$
$$= -\frac{10}{3}$$

So, we have, $g(x) = 3\left(x - \frac{4}{3}\right)^2 - \frac{10}{3}$.

Alternatively,

$$g(x) = 3x^2 - 8x + 2$$
$$= 3\left(x^2 - \frac{8}{3}x\right) + 2$$
$$= 3\left(x^2 - \frac{8}{3}x + \frac{16}{9}\right) + 2 - 3\left(\frac{16}{9}\right)$$
$$= 3\left(x - \frac{4}{3}\right)^2 + 2 - \frac{16}{3}$$
$$= 3\left(x - \frac{4}{3}\right)^2 - \frac{10}{3}$$

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$\therefore g(x) = 3\left(x - \frac{4}{3}\right)^2 - \frac{10}{3}$ which is in the form $a(x+b)^2 + c$,

where $a = 3, b = -\frac{4}{3}$ and $c = -\frac{10}{3}$.

(ii) solve the equation $g(x) = 0$, writing your answer(s) correct to 2 decimal places. [4]

$g(x) = 3\left(x - \frac{4}{3}\right)^2 - \frac{10}{3}$

Let $g(x) = 0$.

$3\left(x - \frac{4}{3}\right)^2 - \frac{10}{3} = 0$

$3\left(x - \frac{4}{3}\right)^2 = \frac{10}{3}$

$\left(x - \frac{4}{3}\right)^2 = \frac{10}{9}$

$x - \frac{4}{3} = \pm\sqrt{\frac{10}{9}}$

$x = \frac{4}{3} \pm \sqrt{\frac{10}{9}}$

Either $x = \frac{4}{3} - \sqrt{\frac{10}{9}}$ or $x = \frac{4}{3} + \sqrt{\frac{10}{9}}$

$x = 0.28$ (to 2 d.p.) $x = 2.39$ (to 2 d.p.)

$\therefore x = 0.28$ or $x = 2.39$

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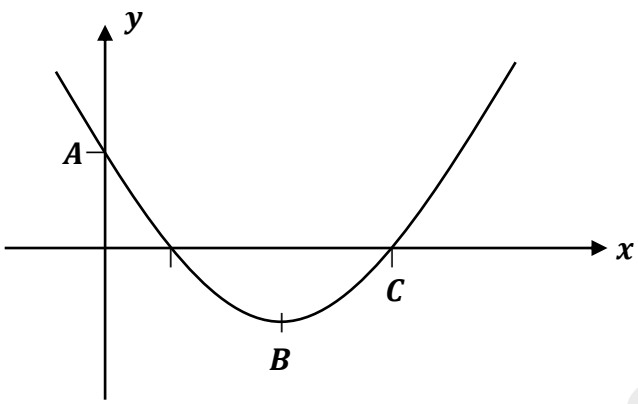
(iii) A sketch of the graph of $g(x)$ is shown below.



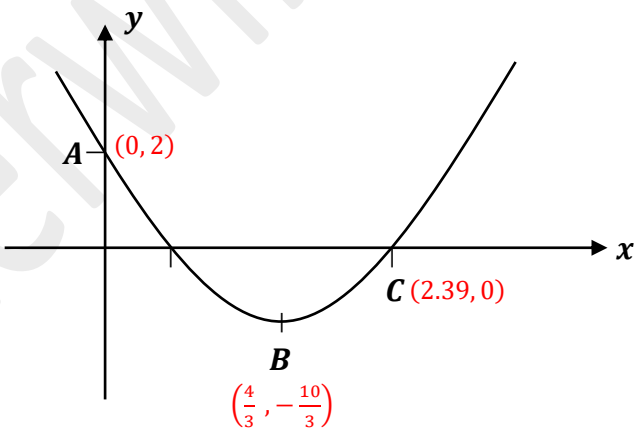
Copy the sketch and state

(a) the y-coordinate of A

(b) the x-coordinate of C

(c) the x and y-coordinates of B [3]

The sketch is shown below:



Total: 15 marks

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10. (a) The manager of a pizza shop wishes to make x small pizzas and y large pizzas. His oven holds no more than 20 pizzas.

(i) Write an inequality to represent the given condition. [2]

Condition: The manager of a pizza shop wishes to make x small pizzas and y large pizzas. His oven holds no more than 20 pizzas.

Inequality: $x + y \leq 20$

(ii) The ingredients for each small pizza cost \$15 and for each large pizza \$30. The manager plans to spend no more than \$450 on ingredients. Write an inequality to represent this condition. [2]

Condition: The ingredients for each small pizza cost \$15 and for each large pizza \$30. The manager plans to spend no more than \$450 on ingredients.

Inequality:

$15x + 30y \leq 450$

$(\div 15)$

$x + 2y \leq 30$

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(b) (i) **Using a scale of 2 cm on the x-axis to represent 5 small pizzas and 2 cm on the y-axis to represent 5 large pizzas**, draw the graphs of the lines associated with the inequalities at (a)(i) and (a)(ii) above. [4]

Consider $x + y \leq 20$ and $x + 2y \leq 30$.

The equations are $x = y = 20$ and $x + 2y = 30$ respectively.

Consider $x + y = 20$.

When $x = 0, y = 20$.

When $y = 0, x = 20$.

Points to be plotted are $(0, 20)$ and $(20, 0)$.

Consider $x + 2y = 30$.

When $x = 0$,

$0 + 2y = 30$

$2y = 30$

$y = \frac{30}{2}$

$y = 15$

When $y = 0$,

$x + 2(0) = 30$

$x + 0 = 30$

$x = 30$

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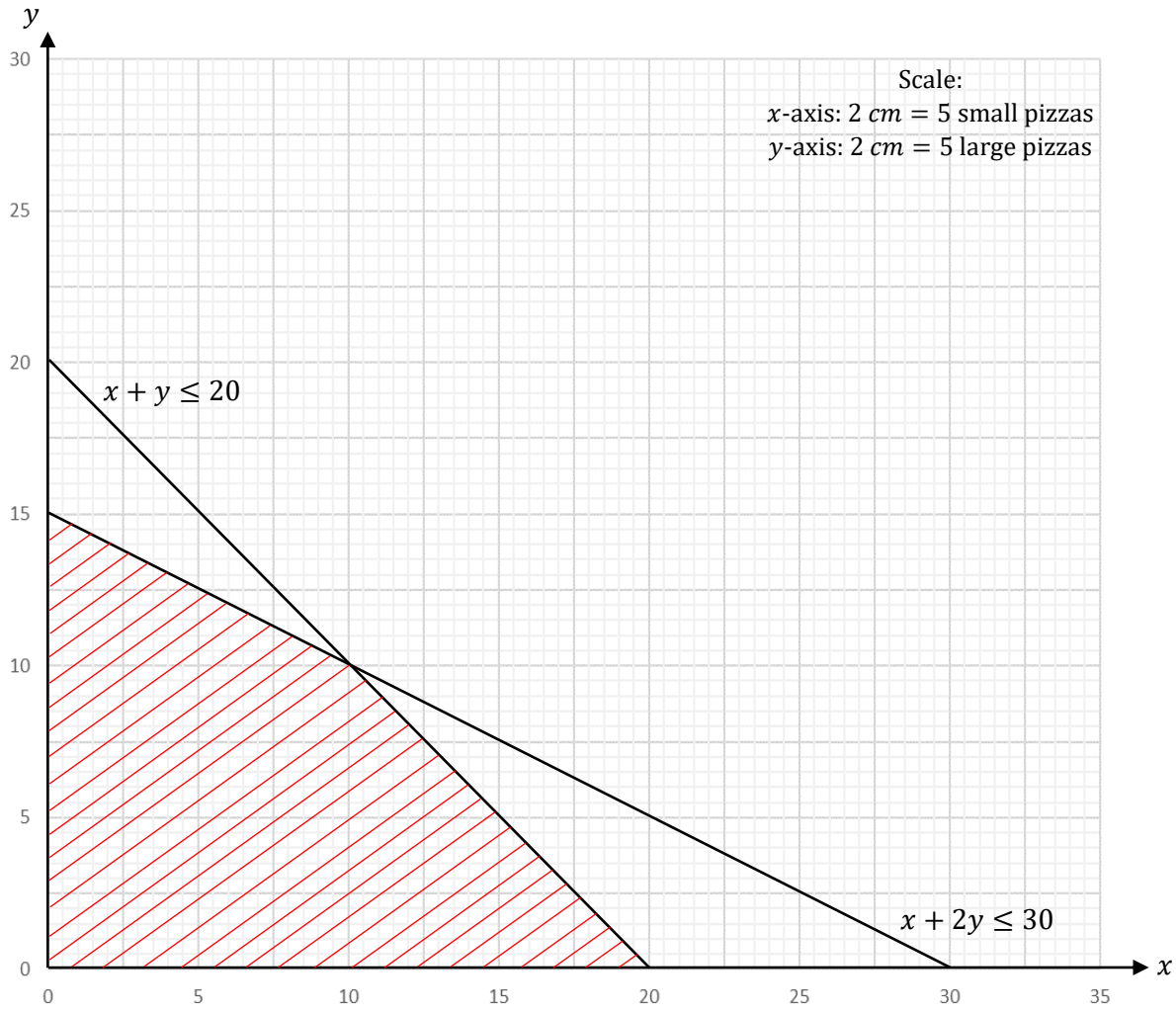
Points to be plotted are $(0, 15)$ and $(30, 0)$.

The graph is shown below:



(ii) Shade the region which is defined by ALL of the following combined:

- the inequalities written at (a)(i) and (a)(ii)
- the inequalities $x \geq 0$ and $y \geq 0$ [1]

See graph above.

(iii) Using your graph, state the coordinates of the vertices of the shaded region. [2]

From the graph, the coordinates of the vertices are:

$(0, 0)$, $(0, 15)$, $(10, 10)$ and $(20, 0)$

(c) The pizza shop makes a profit of \$8 on the sale of EACH small pizza and \$20 on the sale of EACH large pizza. All the pizzas that were made were sold.

(i) Write an expression in x and y for the TOTAL profit made in the sale of the pizzas. [1]

Let the total profit be P.

The profit made on each small pizza is \$8.

The profit made on each large pizza is \$20.

$\therefore P = 8x + 20y$

(ii) Use the coordinates of the vertices given at (b)(iii) to determine the MAXIMUM profit. [3]

The coordinates of the vertices are $(0, 0)$, $(0, 15)$, $(10, 10)$ and $(20, 0)$.

When $x = 0$ and $y = 15$,

$P = 8(0) + 20(15)$

$= 0 + 300$

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$= \$300$

When $x = 10$ and $y = 10$,

$P = 8(10) + 20(10)$

$= 80 + 200$

$= \$280$

When $x = 20$ and $y = 0$,

$P = 8(20) + 20(0)$

$= 160 + 0$

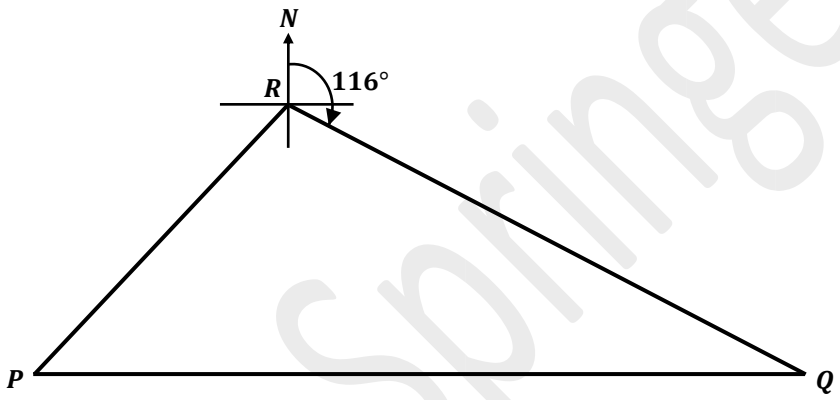
$= \$160$

$\therefore$ The maximum profit occurs when the shop sells 15 large pizzas.

Total: 15 marks

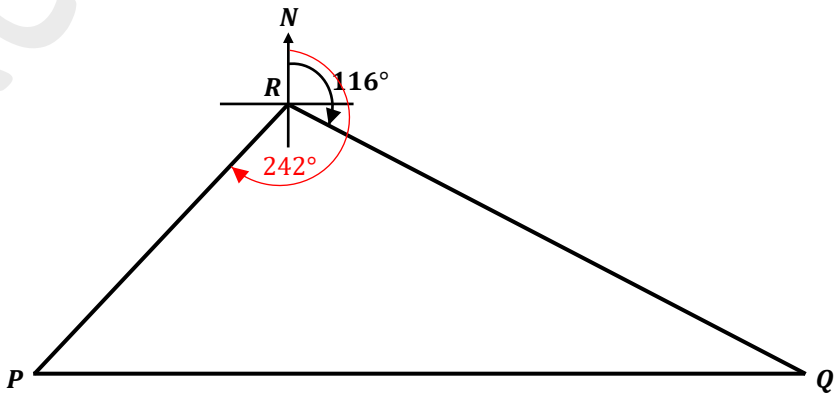
Kerwin Springer

11. (a) The diagram below, **not drawn to scale**, shows three stations P, Q and R, such that the bearing of Q from R is 116° and the bearing of P from R is 242°. The vertical line at R shows the North direction.



(i) Show that angle $PRQ = 126°$. [2]

As stated in the question, 242° was the bearing of P from R as shown below.



242° was the bearing of P from R as shown above.

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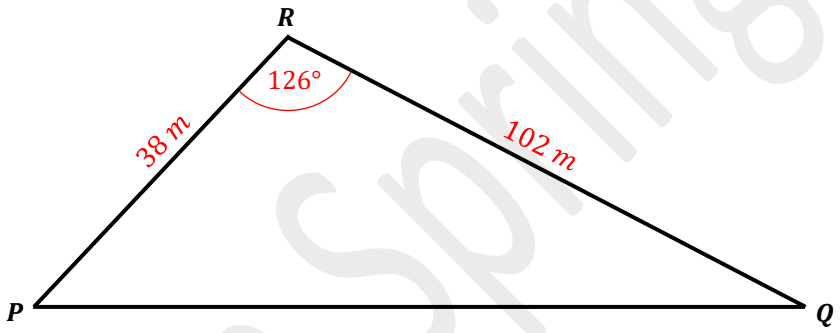
$\therefore P\hat{R}Q = 242° - 116°$

$= 126°$

(ii) Given that $PR = 38$ metres and $QR = 102$ metres, calculate the distance PQ, giving your answer to the nearest metre. [3]

Consider the diagram below:



Using the cosine rule,

$(PQ)^2 = (PR)^2 + (RQ)^2 - 2(PR)(RQ)\cos\hat{R}$

$(PQ)^2 = (38)^2 + (102)^2 - 2(38)(102)\cos 126°$

$(PQ)^2 = 16404.51128$

$PQ = \sqrt{16404.51128}$

$PQ = 128\ m$ (to the nearest metre)

$\therefore$ The distance PQ is $128\ m$.

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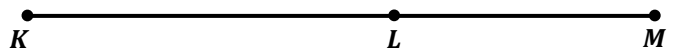
(b) K, L and M are points along a straight line on a horizontal plane, as shown below.

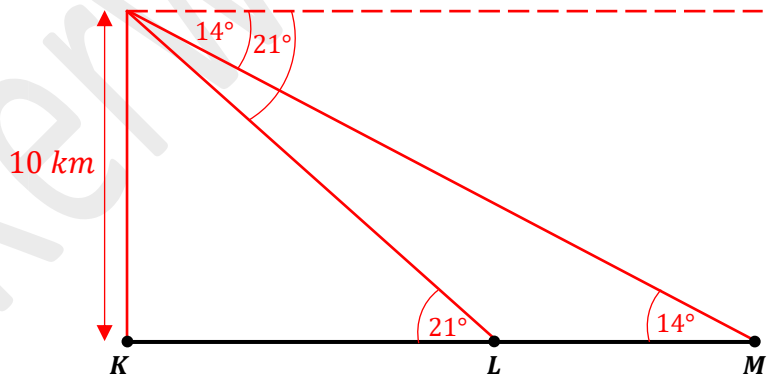


A vertical pole, SK, is positioned such that the angles of elevation of the top of the pole S from L and M are 21° and 14° respectively.

The height of the pole, SK, is 10 metres.

(i) Copy and complete the diagram to show the pole SK and the angles of elevation of S from L to M. [4]

The completed diagram is shown below:



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(ii) Calculate, correct to ONE decimal place,

(a) the length of KL

$$\tan 21° = \frac{10}{KL}$$

$$KL = \frac{10}{\tan 21°}$$

$$KL = 26.1\ m \text{ (to 1 decimal place)}$$

$\therefore$ The length of KL is $26.1\ m$.

(b) the length of LM [6]

$$\tan 14° = \frac{10}{KM}$$

$$KM = \frac{10}{\tan 14°}$$

$$KM = 40.10780934 \text{ m}$$

$$LM = KM - Kl$$

$$= 40.10780934 - 26.1$$

$$= 14.0\ m \text{ (to 1 decimal place)}$$

$\therefore$ The length of LM is $14.0\ m$.

Total: 15 marks

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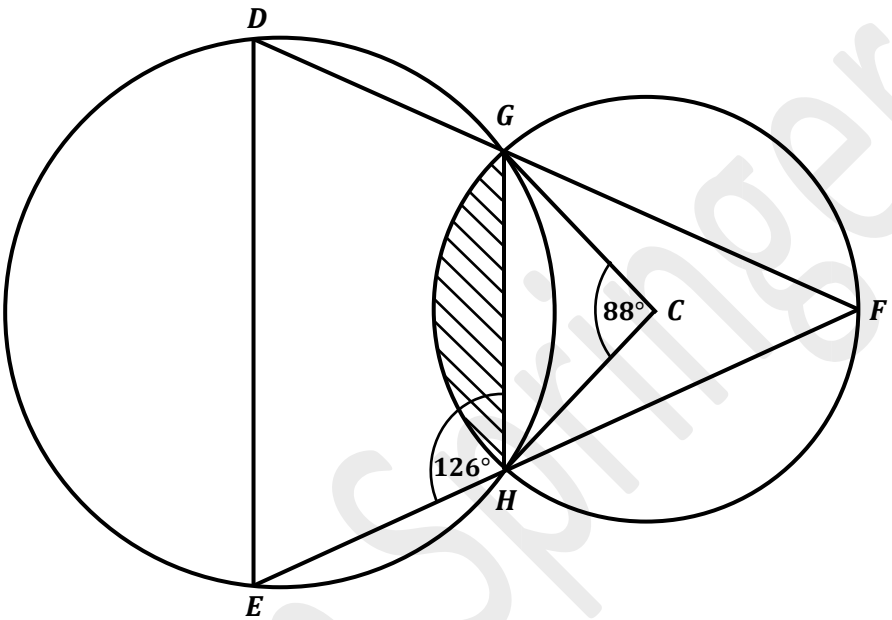
12. (a) The diagram below, **not drawn to scale**, shows two circles. C is the centre of the smaller circle, GH is a common chord and DEF is a triangle.

Angle $GCH = 88°$ and angle $GHE = 126°$



Calculate, giving reasons for your answer, the measure of angle

(i) GFH [2]

The angle subtended by a chord GH at the centre of the circle, C, is twice the angle at the circumference.

Angle $GFH = \frac{1}{2} \times$ Angle GCH

$= \frac{1}{2}(88°)$

$= 44°$

$\therefore$ Angle $GFH = 44°$

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(ii) GDE [3]

The opposite angles of a cyclic quadrilateral are supplementary.

Angle $GDE = 180° - \text{Angle } GHE$

$= 180° - 126°$

$= 54°$

$\therefore$ Angle $GDE = 54°$

(iii) DEF [2]

All angles in a triangle add up to 180°.

Angle $DEF = 180° - \text{Angle } GDE - \text{Angle } GFH$

$= 180° - 54° - 44°$

$= 82°$

$\therefore$ Angle $DEF = 82°$

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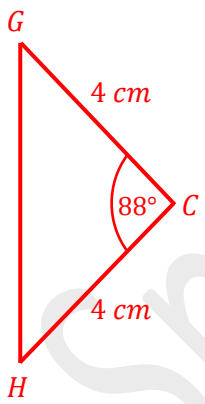
(b) **Use $\pi = 3.14$ in this part of the question.**

Given that $GC = 4\ cm$, calculate the area of

(i) triangle GCH [3]

Since GC and HC are both radii of the same circle, then $GC = HC$.

Consider the diagram below:



Now,

Area of triangle $GCH = \frac{1}{2}ab\sin C$

$= \frac{1}{2}(4)(4)\sin 88°$

$= 7.995\ cm^2$ (to 3 decimal places)

$\therefore$ The area of triangle GCH is $7.995\ cm^2$.

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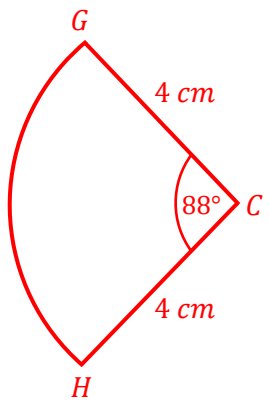
(ii) the minor sector bounded by arc GH and radii GC and HC [3]

Consider the diagram below:



Now,

Area of minor sector $= \frac{\theta}{360°} \times \pi r^2$

$= \frac{88°}{360°} \times 3.14 \times (4)^2$

$= 12.3\ cm^2$ (to 3 significant figures)

$\therefore$ The area of the minor sector is $12.3\ cm^2$.

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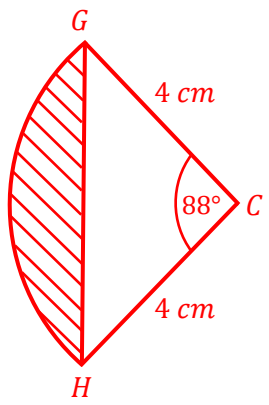
(iii) the shaded segment [2]

Consider the diagram below:



Now,

Area of shaded segment = Area of sector GCH − Area of triangle GCH

$$= \left[\frac{88°}{360°} \times 3.14 \times (4)^2\right] - \left[\frac{1}{2}(4)(4)\sin 88°\right]$$

$$= 4.29\ cm^2 \quad \text{(to 3 significant figures)}$$

∴ The area of the shaded segment is $4.29\ cm^2$.

Total: 15 marks

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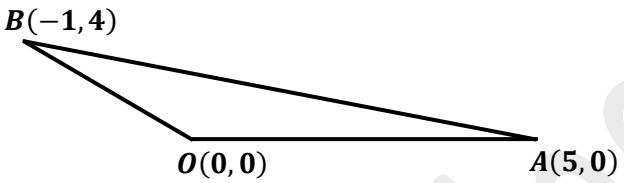
13. (a) The figure below, **not drawn to scale**, shows the points $O(0,0)$, $A(5,0)$ and $B(-1,4)$ which are the vertices of a triangle OAB.



(i) Express in the form $\begin{pmatrix} a \\ b \end{pmatrix}$ the vectors

(a) $\overrightarrow{OB}$

The coordinates of B are $(-1,4)$.

So the vector $\overrightarrow{OB} = \begin{pmatrix} -1 \\ 4 \end{pmatrix}$.

(b) $\overrightarrow{OA} + \overrightarrow{OB}$ [3]

The coordinates of A are $(5,0)$.

So the vector $\overrightarrow{OA} = \begin{pmatrix} 5 \\ 0 \end{pmatrix}$.

Now,

$$\overrightarrow{OA} + \overrightarrow{OB} = \begin{pmatrix} 5 \\ 0 \end{pmatrix} + \begin{pmatrix} -1 \\ 4 \end{pmatrix}$$

$$= \begin{pmatrix} 5 + (-1) \\ 0 + 4 \end{pmatrix}$$

$$= \begin{pmatrix} 4 \\ 4 \end{pmatrix}$$

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(ii) If $M(x, y)$ is the midpoint of AB, determine the values of x and y. [2]

Points are $A(5, 0)$ and $B(-1, 4)$.

Since $M(x, y)$ is the midpoint of AB, then,

$M = \left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$

$= \left(\frac{5+(-1)}{2}, \frac{0+4}{2}\right)$

$= \left(\frac{4}{2}, \frac{4}{2}\right)$

$= (2, 2)$ which is in the form (x, y),

where $x = 2$ and $y = 2$.

$\therefore$ The values of x and y are $x = 2$ and $y = 2$.

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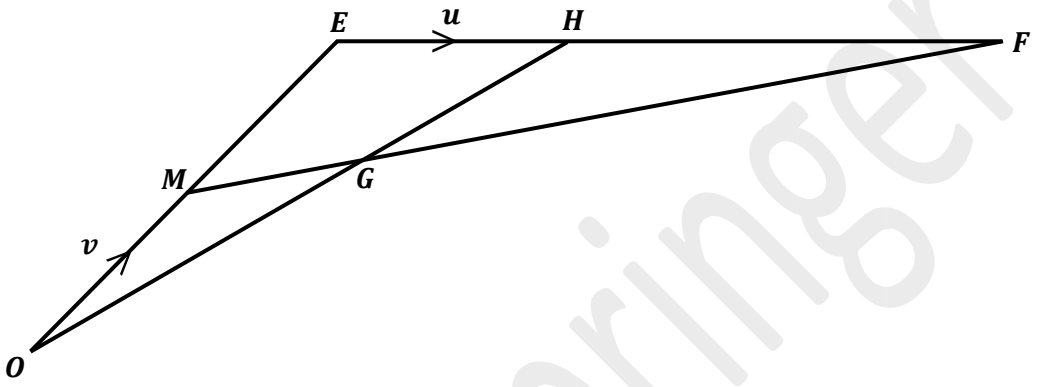
(b) In the figure below, **not drawn to scale**, OE, EF and MF are straight lines. The point H is such that $EF = 3EH$. The point G is such that $MF = 5MG$. M is the midpoint of OE.

The vector $\overrightarrow{OM} = \boldsymbol{v}$ and $\overrightarrow{EH} = \boldsymbol{u}$.



(i) Write in terms of $\boldsymbol{u}$ and/or $\boldsymbol{v}$, an expression for:

(a) $\overrightarrow{HF}$ [1]

We are given that,

$\overrightarrow{EF} = 3\overrightarrow{EH}$

$= 3\boldsymbol{u}$

Now,

Using the triangle law,

$\overrightarrow{EF} = \overrightarrow{EH} + \overrightarrow{HF}$

$\overrightarrow{HF} = \overrightarrow{EF} - \overrightarrow{EH}$

$= 3\boldsymbol{u} - \boldsymbol{u}$

$= 2\boldsymbol{u}$

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(b) $\overrightarrow{MF}$ [2]

Since M is the midpoint of $\overrightarrow{OE}$, then

$\overrightarrow{ME} = \overrightarrow{OM}$

$= \boldsymbol{v}$

Now,

Using the triangle law,

$\overrightarrow{MF} = \overrightarrow{ME} + \overrightarrow{EF}$

$= \boldsymbol{v} + 3\boldsymbol{u}$

(c) $\overrightarrow{OH}$ [2]

Using the triangle law,

$\overrightarrow{OE} = \overrightarrow{OM} + \overrightarrow{ME}$

$= \boldsymbol{v} + \boldsymbol{v}$

$= 2\boldsymbol{v}$

Now,

Using the triangle law,

$\overrightarrow{OH} = \overrightarrow{OE} + \overrightarrow{EH}$

$= 2\boldsymbol{v} + \boldsymbol{u}$

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(ii) Show that $\overrightarrow{OG} = \frac{3}{5}(2v + u)$ [2]

We are given that,

$\overrightarrow{MF} = 5\overrightarrow{MG}$

$= \frac{1}{5}\overrightarrow{MF}$

$= \frac{1}{5}(\boldsymbol{v} + 3\boldsymbol{u})$

Now,

Using the triangle law,

$\overrightarrow{OG} = \overrightarrow{OM} + \overrightarrow{MG}$

$= \boldsymbol{v} + \frac{1}{5}(\boldsymbol{v} + 3\boldsymbol{u})$

$= \boldsymbol{v} + \frac{1}{5}\boldsymbol{v} + \frac{3}{5}\boldsymbol{u}$

$= \frac{6}{5}\boldsymbol{v} + \frac{3}{5}\boldsymbol{u}$

$= \frac{3}{5}(2\boldsymbol{v} + \boldsymbol{u})$

$\therefore \overrightarrow{OG} = \frac{3}{5}(2\boldsymbol{v} + \boldsymbol{u})$

Q.E.D.

(iii) Hence, prove that O, G and H lie on a straight line. [3]

$\overrightarrow{OH} = 2\boldsymbol{v} + \boldsymbol{u}$ and $\overrightarrow{OG} = \frac{3}{5}(2\boldsymbol{v} + \boldsymbol{u})$

So, $\overrightarrow{OG} = \frac{3}{5}\overrightarrow{OH}$

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Since $\overrightarrow{OH}$ is a scalar multiple of $\overrightarrow{OG}$, then $\overrightarrow{OG}$ and $\overrightarrow{OH}$ are parallel.

Since O is a common point on both lines, then G lies on $\overrightarrow{OH}$ (as shown on the diagram).

$\therefore O, G$ and H all lie on a straight line.

Q.E.D.

Total: 15 marks

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14. (a) L and N are two matrices where

$$L = \begin{pmatrix} 3 & 2 \\ 1 & 4 \end{pmatrix} \text{ and } N = \begin{pmatrix} -1 & 3 \\ 0 & 2 \end{pmatrix}.$$

Evaluate $L - N^2$. [3]

$$N^2 = \begin{pmatrix} -1 & 3 \\ 0 & 2 \end{pmatrix}\begin{pmatrix} -1 & 3 \\ 0 & 2 \end{pmatrix}$$

$$= \begin{pmatrix} (-1 \times -1) + (3 \times 0) & (-1 \times 3) + (3 \times 2) \\ (0 \times -1) + (2 \times 0) & (0 \times 3) + (2 \times 2) \end{pmatrix}$$

$$= \begin{pmatrix} 1 + 0 & -3 + 6 \\ 0 + 0 & 0 + 4 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 3 \\ 0 & 4 \end{pmatrix}$$

Now,

$$L - N^2 = \begin{pmatrix} 3 & 2 \\ 1 & 4 \end{pmatrix} - \begin{pmatrix} 1 & 3 \\ 0 & 4 \end{pmatrix}$$

$$= \begin{pmatrix} 2 & -1 \\ 1 & 0 \end{pmatrix}$$

(b) The matrix, M, is given as $M = \begin{pmatrix} x & 12 \\ 3 & x \end{pmatrix}$. Calculate the values of x for which M is singular. [2]

$$M = \begin{pmatrix} x & 12 \\ 3 & x \end{pmatrix}$$

$$\det(M) = ad - bc$$

$$= (x)(x) - (12)(3)$$

$$= x^2 - 36$$

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If M is singular, then the $\det(M) = 0$.

So, we have,

$$x^2 - 36 = 0$$

$$(x+6)(x-6) = 0$$

Either $\quad x + 6 = 0 \quad$ or $\quad x - 6 = 0$

$$x = -6 \qquad\qquad x = 6$$

$\therefore$ The values of x for which M is singular are $x = -6$ and $x = 6$.

(c) A geometric transformation, R, maps the point $(2, 1)$ onto $(-1, 2)$.

Given that $R = \begin{pmatrix} 0 & p \\ q & 0 \end{pmatrix}$, calculate the values of p and q. [3]

$$\begin{pmatrix} 0 & p \\ q & 0 \end{pmatrix}\begin{pmatrix} 2 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ 2 \end{pmatrix}$$

$$\begin{pmatrix} (0 \times 2) + (p \times 1) \\ (q \times 2) + (0 \times 1) \end{pmatrix} = \begin{pmatrix} -1 \\ 2 \end{pmatrix}$$

$$\begin{pmatrix} 0 + p \\ 2q + 0 \end{pmatrix} = \begin{pmatrix} -1 \\ 2 \end{pmatrix}$$

$$\begin{pmatrix} p \\ 2q \end{pmatrix} = \begin{pmatrix} -1 \\ 2 \end{pmatrix}$$

Comparing corresponding entries gives:

$p = -1 \quad$ and $\quad 2q = 2$

$$q = \frac{2}{2}$$

$$q = 1$$

$\therefore p = -1$ and $q = 1$

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(d) A translation, $T = \begin{pmatrix} r \\ s \end{pmatrix}$ maps the point $(5, 3)$ onto $(1, 1)$. Determine the values of r and s. [3]

$$\begin{pmatrix} r \\ s \end{pmatrix} + \begin{pmatrix} 5 \\ 3 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$\begin{pmatrix} r+5 \\ s+3 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

Comparing corresponding entries gives:

$r + 5 = 1$ and $s + 3 = 1$

$r = 1 - 5$ $\quad$ $s = 1 - 3$

$r = -4$ $\quad$ $s = -2$

$\therefore r = -4$ and $s = -2$

(e) Determine the coordinates of the image of $(8, 5)$ under the combined transformation, R followed by T. [4]

$$R = \begin{pmatrix} 0 & p \\ q & 0 \end{pmatrix}$$

Since $p = -1$ and $q = 1$, then $R = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$.

Firstly, we transform under R.

$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 8 \\ 5 \end{pmatrix} = \begin{pmatrix} (0 \times 8) + (-1 \times 5) \\ (1 \times 8) + (0 \times 5) \end{pmatrix}$$

$$= \begin{pmatrix} 0 + (-5) \\ 8 + 0 \end{pmatrix}$$

$$= \begin{pmatrix} -5 \\ 8 \end{pmatrix}$$

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$T = \begin{pmatrix} r \\ s \end{pmatrix}$

Since $r = -4$ and $s = -2$, then $T = \begin{pmatrix} -4 \\ -2 \end{pmatrix}$.

Then, we transform under T.

$$\begin{pmatrix} -5 \\ 8 \end{pmatrix} + \begin{pmatrix} -4 \\ -2 \end{pmatrix} = \begin{pmatrix} -5 + (-4) \\ 8 + (-2) \end{pmatrix}$$

$$= \begin{pmatrix} -5 - 4 \\ 8 - 2 \end{pmatrix}$$

$$= \begin{pmatrix} -9 \\ 6 \end{pmatrix}$$

$\therefore$ The image of $(8, 5)$ under the combined transformation, R followed by T is $(-9, 6)$.

Total: 15 marks

END OF TEST

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